MINING DEALER BEST PRACTICE

Ladder for Maintenance and R&I of Front Suspension and Steering Wheel Accumulators on 797 truck models

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

To ensure availability of and confidence in equipment, it is necessary to effectively execute maintenance and repair tasks. Such tasks must be performed efficiently, with good quality and safety standards.

In the case of large size mining equipment, such as the 797 truck models, the tasks include, among others, the need to reach several elevated and/or narrow zones to replace components or carry out maintenance activities. To achieve the safety, efficiency, and quality goals, the maintenance department must ensure its staff has the needed skills, support information (technical and procedures), infrastructure, tools, and support.

This Best Practice is about the design, making, use, and benefits of a tool developed to help in the maintenance and R&I of front suspension and steering wheel accumulators on 797 truck models, in order to be able to reach the goals mentioned earlier.

2.0 Best Practice Description

2.1 Opportunity Detection

The current procedure for maintenance and R&I of front suspension and steering wheel accumulators in the Caterpillar service manual requires that the maintenance team perform tasks at height (above 2 meters). These tasks are carried out with a portable ladder and two technicians who need to coordinate with each other safety measures to avoid accidents due to horizontal and vertical movements of the ladder.

This Best Practice proposes a ladder designed to be used by only one technician, obtaining a better safety standard and avoiding the risk of movement or tilt of the ladder.

Using this ladder under our DMH contract has allowed the identification of three (3) large areas with improvement opportunities.

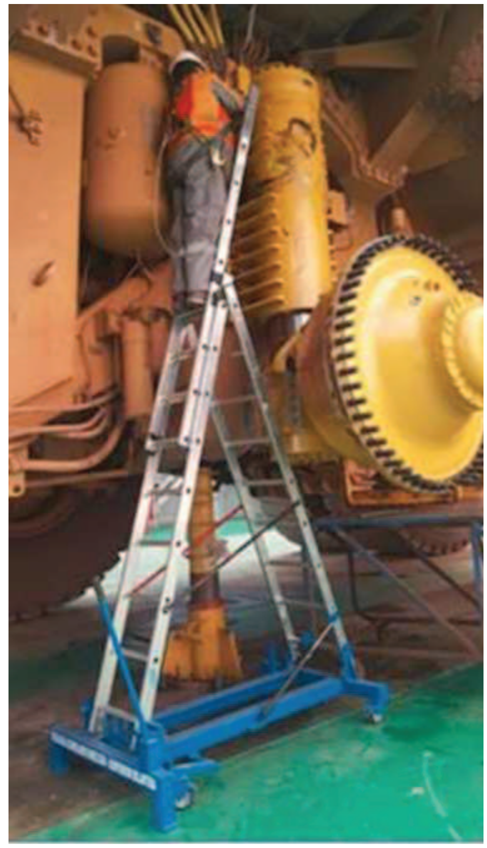
<u>Improvement Opportunities</u>

Safety

A task at height is considered a high-risk type, therefore our customers establish high standard safety policies.

Using this tool reduces considerably the risks of horizontal and vertical movement, decreasing the risk of fall accidents from a raised position.

Before



Now

Resources (Labor)

The original procedure for this job requires the use of two (2) maintenance keepers. One of them holds the ladder permanently to avoid horizontal and vertical movements, while the other performs the specific task. This ladder has a mechanism to keep it fixed permanently.

Quality

Tasks can be performed smoothly without distractions on this ladder due to its fixed base, which results in a more efficient and effective work.

2.2 Proposed Solution of Improvement

The solution proposed is the use of an expandable tool ladder, especially designed to perform activities of Maintenance and R&I of front suspension and steering wheel accumulators **ESDAD-DMH** (**F**ront **S**uspension **L**adder and **S**teering **W**heel **A**ccumulators - **D**ivision **M**inistro **H**ales).

Expandable ladder

This Tool allows for a simple, highly maneuverable handling, and it is safe to perform tasks within scheduled times.

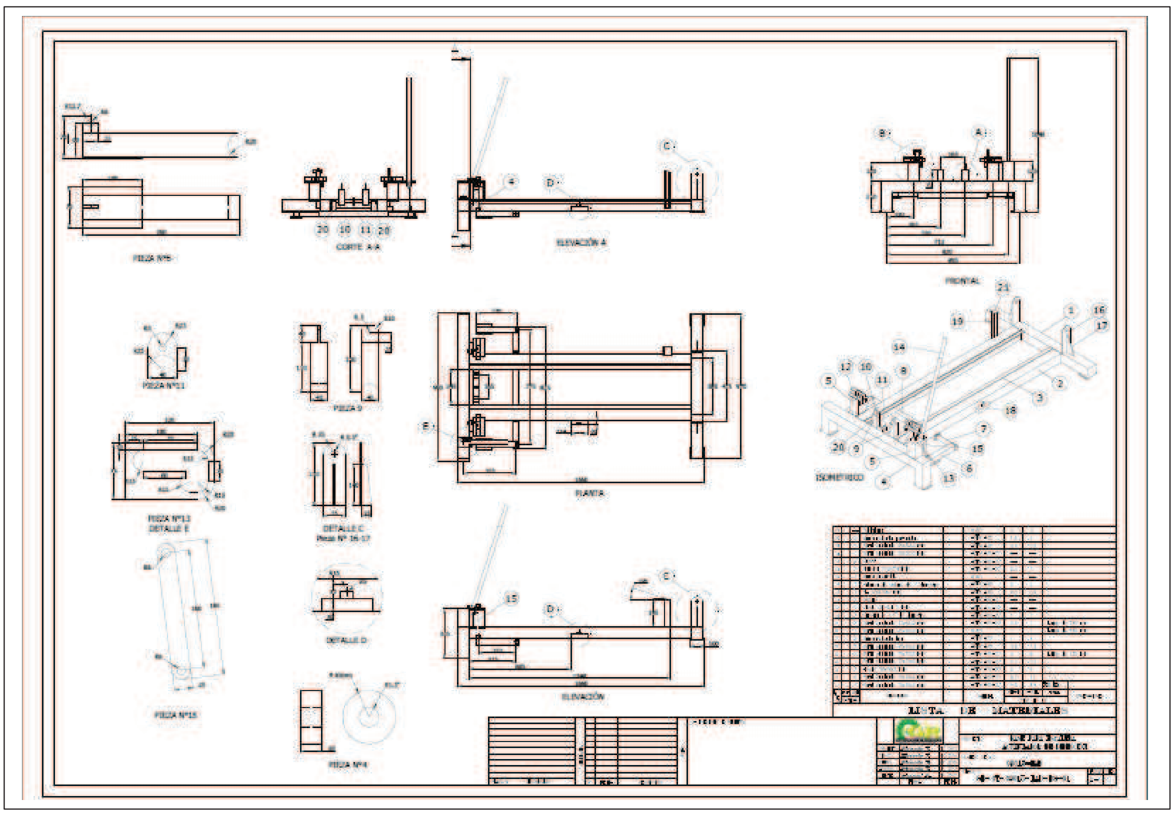
3.0 Steps for Implementation

The general steps to follow in the implementation of this Best Practice are:

3.1 Manufacturing the ESDAD-DMH
3.2 Technical certification of the ESDAD-DMH
3.3 Practical tests on tasks of Maintenance and R&I of Front Suspension and steering wheel accumulator
3.4 Final adjustments
3.5 Documentation of SOP (Standard Operation Procedure)
3.6 Staff training
3.7 Operational Start-up and Validation of Benefits

3.1 Manufacturing the ESDAD-DMH

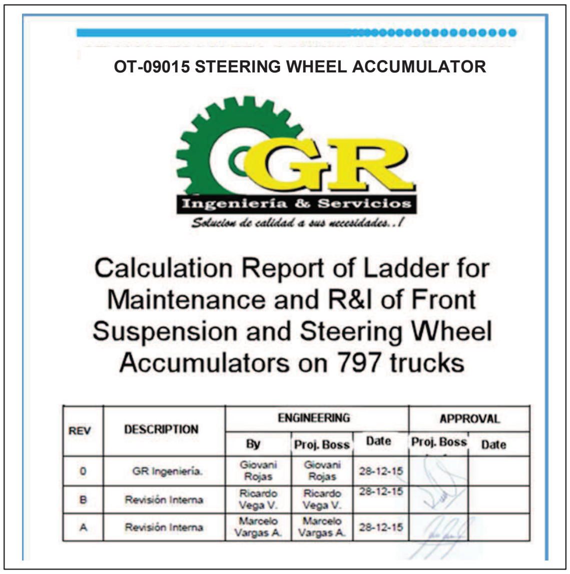
The ESDAD-DMH was locally manufactured, following these particular design specifications. The design was made locally under the Finning - DMH contract.



This manufacturing plan is attached to this Best Practice.

3.2 Technical Certification of Maintenance Tool and R&I of Suspensions and Steering Wheel Accumulators (Welds and Capacity)

The certification of the welds and capacity analysis of the accessory is important once manufactured


OT-09015 STEERING WHEEL ACCUMULATOR

GR Ingeniería & Servicios

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Calculation Report of Ladder for Maintenance and R&I of Front Suspension and Steering Wheel Accumulators on 797 trucks

REV	DESCRIPTION	ENGINEERING			APPROVAL	
		By	Proj. Boss	Date	Proj. Boss	Date
0	GR Ingeniería.	Giovani Rojas	Giovani Rojas	28-12-15		
B	Revisión Interna	Ricardo Vega V.	Ricardo Vega V.	28-12-15		
A	Revisión Interna	Marcelo Vargas A.	Marcelo Vargas A.	28-12-15		

Structural Calculation Report and Quality Certification of ESDAD-DMH

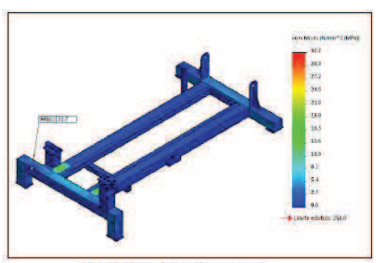

Figura 7.3. Distribución de esfuerzo en elemento. (Esfuerzo Máx 32.7 MPa).

Testing samples / stress analysis made on ESDAD-DMH

3.3 Practical tests on tasks of Maintenance and R&I of front suspension and accumulators

Once the Tool is received we recommend operational testing at the workshop in a safe and controlled environment, in order to identify and provide the final adjustments required.

3.4 Final adjustments

Adjust as required and detected.

3.5 Documentation of SOP (Standard Operation Procedure)

In order to ensure the correct operation of this tool, we recommend documenting the standard operation procedure of use (SOP).

Additionally, we recommend checking the specific safety regulations for its operation in order to adapt the SOP to its specific needs.

3.6 Staff Training

The staff must be adequately trained on the daily use of the ESDAD-DMH for Maintenance tasks and R&I of front suspension and steering wheel accumulators on 797 truck models.

It is recommended that this training include the revision of the standard procedure (SOP) as well as the practice of use in a safe and controlled environment.

3.7 Operational Start-up and Validation of Benefits

Once the ESDAD-DMH is under operation, a follow-up of the correct use must be carried out along with a record of the benefits.

Once the expected benefits are obtained, the specific Standard Jobs must be improved/modified with the Planning & Scheduling department, especially the new halted machine times and man-hour.

4.0 Benefits

According to our observations and records of use of this accessory in our operation, the documented benefits so far show the following improvements.

Change of components and maintenance of front suspension and steering wheel accumulators.

Task 797 Truck		Risk Level	Technicians	Time required [hr]	Total MH [hr]
Checking of Front Suspension load	No ESDAD	Medium	2	3 <> 6	6<>12
	With ESDAD	Low Controlled	1	3<>6	3<>6
Checking of Steering Wheel Accumulator	No ESDAD	Medium	2	1	2
	With ESDAD	Low Controlled	1	1	1

As shown in the table above, the benefits can be seen in the areas of Safety, as well as an optimization of Labor.

Safety: The change is from human support to mechanical support of the ladder, which significantly reduces the levels of risk.

Man-Hours (MH used): The reduction of 1 support person to this task means having more man-hours of availability, which can be assigned to other tasks.
The reduction of MH used for this task will positively influence the results of MR (Maintenance Ratio) of the fleet.

Quality: While it is difficult to quantify, we have detected improvements in relation to the quality of operation when replacing components and maintenance of front suspension and steering wheel accumulators on 797 truck models due to an improved visibility of the areas which are difficult to check.

5.0 Required Resources

The required resources are specified in this Best Practice

6.0 Additional Support / References

The following information is attached to this Best Practice

- Diagram / Manufacturing Plan

7.0 Other Related Best Practices

N/A

8.0 Acknowledgements

Creation, Implementation, and Documentation of the Best Practice

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